Boundary-Induced Collapse (BIC) in Spectral Language

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1. System–boundary structure

Let a quantum system S interact with a boundary B (environment, apparatus, or constraint structure).
The total Hamiltonian is

$$\hat{H}_{\text{tot}} = \hat{H}_S + \hat{H}_B + \hat{H}_{SB}.$$

The effective dynamics of S is governed by an induced operator

$$\hat{G} = \mathrm{Tr}_B\Big(\hat{H}_{SB}\, \rho_B\, \hat{H}_{SB}^{\dagger}\Big),$$

which is Hermitian and positive semidefinite.

This operator encodes the **boundary constraints** acting on the system.

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2. Spectral decomposition of the boundary operator

By the spectral theorem,

$$\hat{G} = \sum_i \lambda_i |\phi_i\rangle\langle\phi_i|, \qquad \lambda_i \geq 0.$$

The eigenstates $|\phi_i\rangle$ define the **spectral boundary basis**.

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3. BIC Postulate (Spectral Form)

Boundary-Induced Collapse Postulate (Spectral).

> *A quantum system subject to boundary constraints collapses into the spectral basis of the induced boundary operator $\hat{G}$. The eigenstates $|\phi_i\rangle$ are ontological deltas, and the eigenvalues λ_i quantify their structural necessity.*

4. Action on the density matrix

Let the system density matrix be ρ.
Under BIC, its evolution satisfies

$$\rho \longrightarrow \sum_i \Pi_i \rho \Pi_i, \qquad \Pi_i = |\phi_i\rangle\langle\phi_i|.$$

This is exactly the spectral projection induced by $\hat{G}$.

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5. Relation to decoherence

Environmental decoherence corresponds to the dynamical suppression of off-diagonal terms in this same spectral basis:

$$\rho \to \sum_i p_i |\phi_i\rangle\langle\phi_i|.$$

BIC identifies this basis as **boundary-determined**, not arbitrary.

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6. Collapse as spectral stabilization

In BIC, collapse is not a discontinuous postulate but a **spectral stabilization**:

$$\rho(t) \xrightarrow{\text{BIC}} \rho_{\text{diag}} = \sum_i p_i |\phi_i\rangle\langle\phi_i|.$$

Outcome selection corresponds to dominance of one stabilized spectral delta under boundary constraints.

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7. Connection with linear spectral systems

Your linear method:

$$A^T A = Q D Q^T$$

is the classical analog of

$$\hat{G} = \sum_i \lambda_i |\phi_i\rangle\langle\phi_i|.$$

Both represent:

> *Boundary-induced diagonalization of structural degrees of freedom.*

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8. Ontological interpretation (TNA)

SPECTRAL OBJECT	TNA MEANING
$\|\phi_i\rangle$	Eigenstate (Spectral Delta)
λ_i	Ontological weight
Spectral projection	Collapse
Boundary operator $\hat{G}$	Structural necessity operator
Diagonal density matrix	Reduced incoherence state

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9. Compact BIC spectral law

$$\boxed{\text{Boundary} \;\Rightarrow\; \hat{G} \;\Rightarrow\; \text{Spectral deltas} \;\Rightarrow\; \text{Collapse.}}$$

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10. BIC Theorem (Spectral Form)

Theorem (BIC Spectral Collapse).

Let $\hat{G}$ be the Hermitian operator induced by boundary interaction. Then the collapse dynamics of the system is governed by the spectral decomposition of $\hat{G}$, and the observable outcomes correspond to its eigenstates.

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11. Measurement as special case

If $\hat{G} = \hat{M}$ is a measurement operator, BIC reduces to standard projective measurement.

Thus:

> *Quantum measurement is a particular case of boundary-induced spectral collapse.*

12. Final ontological statement

> *Boundary-Induced Collapse is the physical realization of spectral necessity.*

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13. Closing sentence for paper

> *In spectral language, Boundary-Induced Collapse is the stabilization of ontological deltas defined by the eigenstructure of the boundary operator, unifying decoherence, measurement, and structural necessity within a single mathematical framework.*